# Ramp energy lab

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Compare gravitational energy change with measured kinetic energy.

**SUCCESS CRITERIA - I can...**

1. Measure height vertically and speed at a stated location.
2. Calculate an energy balance.
3. Discuss rotation, resistance and uncertainty.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

An ideal system has total mechanical energy 15 J. At a point where U=3 J, find K.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $mg\Delta h = \Delta K_{translation} + \Delta K_{rotation} + \Delta E_{thermal}$. Closed energy accounting near Earth; Δh is the positive drop. Determine whether the object rolls or slides; use the approved ramp setup.

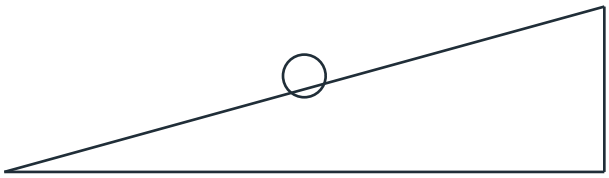
**Worked reasoning:** A supplied ramp run loses 6 J of gravitational potential energy but gains only 4 J of translational kinetic energy. The 2 J difference needs an account, such as rotation or friction.

**Your turn:** Draw an energy account with the 2 J remainder explicitly labeled as not yet identified.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

On the ramp sketch, mark the vertical height h and the along-ramp path. Which belongs in mgh?



Label h vertically. The marker represents the object on the sloping surface.

**Marzano check - circle:** ① ② ③

## 04 Build the solution

Supplied model: mass 1 kg drops .50 m; g=10. At bottom translational K=4 J. Find lost U and energy not accounted for by translation alone.

**Marzano check - circle:** ① ② ③

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## 05 Use the idea

Create a table for repeated trials. Include actual height, mass, bottom speed,ΔU, K and one observation about losses or rotation.

**Marzano check - circle:** 1 2 3

## 06 Choose a method

Keep mass fixed and multiply speed by 4. By what factor does kinetic energy change? Explain from the equation.

**Marzano check - circle:** 1 2 3

## 07 Exit check

A rolling object has less½mv² than its lost mgh. Name two possible destinations of the difference.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

Design a comparison that helps distinguish rolling energy from sliding friction as explanations for a lower translational speed.

**Marzano check - circle:** 1 2 3